Pengchen Zong (宗鹏陈)



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Affiliation: School of Artificial Intelligence, Xi'an Jiaotong University
Advisors: Prof. Pengju Ren and Prof. Tian Xia

Education

- 2023.09–present | **Ph.D. Candidate, Artificial Intelligence** | Xi'an Jiaotong University, Xi'an
  Expected defense: February 2028. Co-advised by Prof. Pengju Ren and Prof. Tian Xia.
- 2018.09–2021.06 | **M.S., Electronic Science and Technology** | Xi'an Jiaotong University, Xi'an
  Rank 9/93. Outstanding Graduate Student; First-Class Academic Scholarship. Co-advised by Prof. Pengju Ren and Prof. Tian Xia.
- 2014.09–2018.06 | **B.S., Microelectronics Science and Engineering** | Northwestern Polytechnical University, Xi'an

Research Interests

My interests span computer architecture, hardware–software co-design, and memory systems, particularly prefetching for irregular workloads. I am now exploring LLM-agent execution on CPU–GPU systems, where explicit plans may expose future data and execution-state needs, but uncertain branches require runtime feedback to balance latency gains against wasted bandwidth, memory capacity, and compute.

Publications

1. **Pengchen Zong**, Gelin Fu, Qiang Luo, Yijiang Kong, Chenzhun Guo, Wenzhe Zhao, Pengju Ren, and Tian Xia. "Loop-Decoupled Prefetcher for Linked Data Structure." *MICRO*, 2026. Accepted.

2. **Pengchen Zong**, Tian Xia, Haoran Zhao, Jianming Tong, Zehua Li, Wenzhe Zhao, Nanning Zheng, and Pengju Ren. "PIT: Processing-In-Transmission with Fine-Grained Data Manipulation Networks." *IEEE Transactions on Computers*, 2020.

3. Tian Xia, **Pengchen Zong** (First Student Author), Haoran Zhao, Jianming Tong, Wenzhe Zhao, Nanning Zheng, and Pengju Ren. "COCOA: Content-Oriented Configurable Architecture Based on Highly-Adaptive Data Transmission Networks." *ACM GLSVLSI*, 2020.

4. **Pengchen Zong**, Yijiang Kong, Yuwen Mao, Guoming Yang, Jiahao Pan, Gelin Fu, Wenzhe Zhao, Pengju Ren, and Tian Xia. "POP: Program-Order Prefetcher for Indirect Memory Access." *DATE*, 2027. Under review.

5. Gelin Fu, Tian Xia, **Pengchen Zong** (First Student Author), Qiang Luo, Endian Li, and Pengju Ren. "LA-DMP: Loop-Aware Differential Matching Prefetcher for Indirect Memory Access." *ACM TACO*. Under review.

Industry Experience

- 2021.07–2023.08 | **Digital IC Design Engineer** | HiSilicon Technologies, Shanghai
  WiFi-SoC: led the on-chip debug and observability architecture; integrated an AI compute core; SoC bus integration and verification.
- 2020.07–2020.09 | **Intern, AI Hardware** | Alibaba T-Head / DAMO Academy, Shanghai
  Fully associative instruction cache with data backpressure: specification, RTL, verification, and synthesis.

Awards

1. DAC 2019 System Design Contest, FPGA track, 2nd place worldwide. HiPU-100 on Ultra96 running ShuffleNet-V2.

2. ICCV 2019 Low-Power Image Recognition Challenge, 3rd place worldwide. EfficientNet-B4 on Ultra96.

3. First China Graduate AI Innovation Competition (Intel Cup), national first prize. HiPU accelerator; cache subsystem and live demonstration.